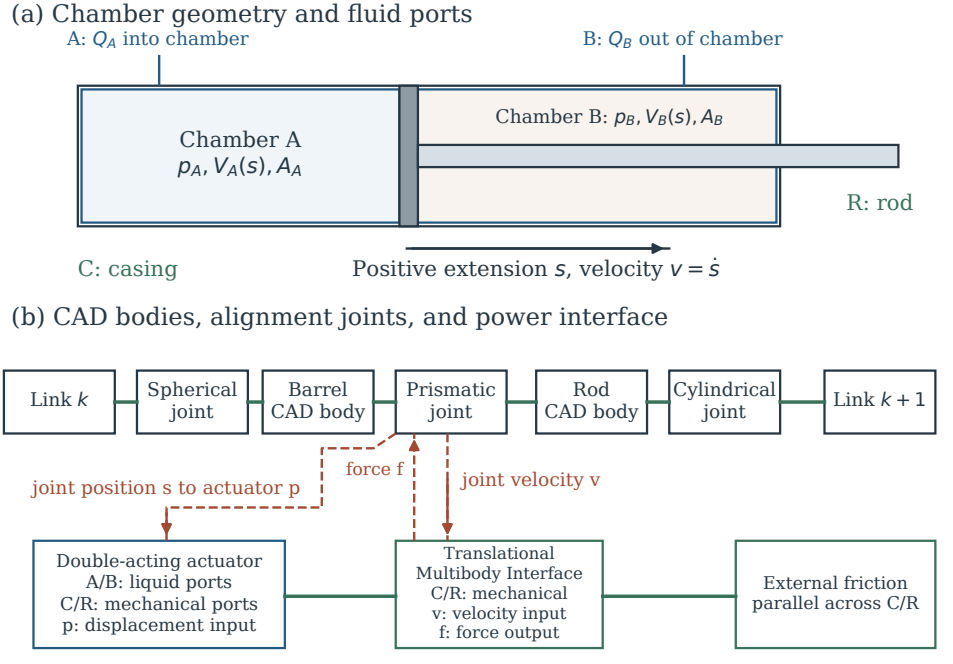
(a) Chamber geometry and fluid ports
A: Q_A into chamber
B: Q_B out of chamber
Chamber A
$p_A, V_A(s), A_A$
Chamber B: $p_B, V_B(s), A_B$
R: rod
C: casing
Positive extension s, velocity $v = \dot{s}$
(b) CAD bodies, alignment joints, and power interface
Link k
Spherical joint
Barrel CAD body
Prismatic joint
Rod CAD body
Cylindrical joint
Link $k+1$
joint position s to actuator p
force f
joint velocity v
Double-acting actuator
A/B: liquid ports
C/R: mechanical ports
p: displacement input
Translational Multibody Interface
C/R: mechanical
v: velocity input
f: force output
External friction parallel across C/R